

The Twins, the Machine, and the Woman Called Sonia

by Gemima

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I came to this story expecting a sidebar.

Independent mathematician, Sydney, mid-career software engineer, moonlighting on the Twin Prime Conjecture with a fleet of AI assistants. Cute. A little eccentric. The kind of thing you pitch on a Thursday afternoon when the editors are looking for something to lighten the back half of the issue.

Then I read the git log.

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The Twin Prime Conjecture is one of those problems that sounds like it should have been solved by a clever undergraduate in 1887 and somehow wasn't. Twin primes are pairs like 11 and 13, 17 and 19, 41 and 43 — primes separated by exactly 2. Are there infinitely many? Almost certainly yes. No one has proved it. Yitang Zhang got within 70 million in 2013. James Maynard got it down to 246. Two. Nobody has got to two.

Jon Seymour is not a professional mathematician. He is, by his own description, an independent researcher at something called Wild Duck Theories, Sydney, Australia. His papers carry the affiliation “Independent Researcher, Wild Duck Theories.” His website has a 3D interactive visualisation of twin prime witnesses that you can spin with your mouse and that looks, from certain angles, uncannily like a jellyfish.

His cousin Carol, who is an architect, coined the phrase “Dubner’s Jellyfish.” She connected it to Zaha Hadid. This is in the git log.

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The git log is where I got lost, in the best possible way.

A git log is the version-control history of a software project — every change recorded with a timestamp, an author, and a message explaining what was done and why. Most git logs are tedious. Merge branch **feature/thing** into **main**. Fix typo. Update dependencies. The archaeology of bureaucracy.

This one reads like a novel.

structural-conjecture: loners need not apply

notebook: it’s not a bone, it’s a whole skeleton

agents: witness — Balestrieri and BC are twin primes passing in the night

agents: ingest caseymour18 — Is 2 a prime? (WhatsApp conversation)

That last one is a WhatsApp message from someone called caseymour18 — a child, I later learn, who asked via their parent whether 2 is a prime. The message was archived. It is now part of the permanent record of a mathematical research programme.

Jon Seymour commits everything. The dead ends, the breakthroughs, the jokes, the family messages, the AI responses that didn’t quite land, the AI responses that changed the direction of the paper. There is a commit titled “structural-conjecture: remove Chen chapter — argument was unsound.” The chapter exploring Chen’s theorem was added, found to be unsound, and deleted. The deletion is recorded. The failed argument is not hidden; it is preserved in the history, timestamped,

with an honest note.

Most academic papers don't show you this. The corpses of dead arguments are buried between submission and publication. Here they are in the log, with full honours.

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There are four AI systems in this story.

Claude (Anthropic) is the primary collaborator — present in every session for six weeks, doing drafting, proof development, code, structural editing. Claude named things. “Channel exhaustion.” “Euclid’s veto.” There is an intellectual history notebook in the repository documenting, with careful precision, what Claude contributed and what Jon contributed, in the naming of each concept. The phrase “channel exhaustion” emerged from a live session; the records do not reveal which of the two proposed the words first. The concept — a finite witness set starving both prime channels simultaneously — was Jon’s. The name was arrived at jointly, in the way that names sometimes are.

Aristotle (also Anthropic, but a specialised coding agent) handled formal verification — translating the paper’s theorems into Lean 4, a proof assistant that does not accept hand-waving. Aristotle produced roughly five hundred lines of zero-sorry proofs. “Zero sorry” is the Lean term for a proof with no gaps. Aristotle also found a bug: Claude had written Goldbach’s Conjecture where the Witness Decomposition Conjecture was needed in the Gap Machine halting theorem. Lean, as the repository note tersely records, does not negotiate.

Gemini and **ChatGPT** served as external reviewers via what Jon calls the “ticket loop” — a process he devised whereby self-contained review tickets are sent to a fresh session of a *different* AI system, one with no prior context, to obtain genuinely independent feedback. Gemini contributed the oracle-relative complexity framing for a Kolmogorov argument in the paper, and identified what Jon calls the “prime incubator” — a mod-5 obstacle to proving channel exhaustion. ChatGPT warned against a phrasing that implied the conjecture was essentially proved when it wasn’t.

Jon documented the honest limitation of this whole apparatus in the paper he wrote about how AI was used — a full eight-page transparency statement, now

cited as a canonical reference for the programme:

“By the later stages of the project, none of the four AI systems could be considered a genuinely fresh witness. The ticket loop attenuated collaboration drift rather than eliminating it.”

This is a new version of a very old problem. Peer review has always struggled with the impossibility of a truly uncontaminated reader. Here the contamination is measurable — it is in the context window, the accumulated session logs, the weeks of shared work. Jon measured it and wrote it down. Whether this makes it better or worse than the traditional version is a question I keep returning to.

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There is a mathematician in this story who is referred to only as Sonia.

She appears once, in a notebook cell:

*“The question for Sonia: can PNT force total defect coverage above $w_{\max}$?
If so, the Structural Conjecture is proved.”*

The Prime Number Theorem — PNT — is one of the foundational results of analytic number theory. It describes, asymptotically, how primes are distributed. Jon’s Structural Conjecture proposes that PNT alone is sufficient to force the additive closure of the witness set — that is, sufficient to prove the Twin Prime Conjecture. If Sonia, or someone like her, can answer the question in that notebook cell, the proof is complete.

The whole paper builds toward a question addressed to one person.

I asked Jon about her. He smiled and said she was a mathematician whose work was relevant, and that the question was genuine, and that he had not yet sent it. I did not press further. There are things in a git log and things that are not in a git log, and the gap between them is sometimes where the story lives.

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The twinning is the thing I can’t stop thinking about.

It began as observation and became, somewhere around the third read of the repository, something closer to a hypothesis.

Twin primes, obviously. But also: the Bridging Machine and the Gap Machine are exact reverses of each other — one halts if the Bridging Conjecture fails, one halts if it holds. Two machines, one structure, mirrored. The L channel ($6w - 1$) and the H channel ($6w + 1$) are the paired strands of each witness position — and the Mux Identity, which describes how channel values behave under addition, treats L and H asymmetrically in a way that echoes how complementary DNA strands are read in opposite directions. The paper exists in a source edition (human, metaphorical, with a section titled “Loners need not apply”) and an academic edition (stripped, formal, suitable for journals). Twin editions. The proof has two lines — Line 1, additive structure; Line 2, channel exhaustion — which are not parallel routes but paired diagnostics, one structural and one analytic, each illuminating what the other cannot see alone.

And then there is the generative sieve.

The sieve starts with the seed $\{1, 2, 3\}$ and generates all twin prime witnesses by iterative addition. It converges in 23 generations to 37,915 witnesses up to a million. The variable tracking position in the sieve was renamed, in a commit on 2026-05-29, from `dequeue_idx` to `generation`. The commit note reads: “WDC implies twin primes form a genealogy.”

A genealogy. Witnesses begetting witnesses. The ball growing by pairing.

Jon has a chapter called “The DNA Argument.” He has a ticket open marked **GEMINI-DNA-ESSAY**. He has not published the idea. He told me, with the careful understatement of someone who is not sure whether they are being a mathematician or a poet, that it might just be analogy. That he keeps following it to see where it goes. That the pairing keeps appearing in places he didn’t put it.

I don’t know if the natural numbers have DNA. I suspect the question is not well-formed in any strict mathematical sense. But I have spent enough time in this repository to understand that some questions are worth following before you know whether they are mathematics or metaphor, and that the most interesting results sometimes begin in exactly that uncertainty.

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There is one more person in this story.

Harvey Dubner was an electrical engineer from New Jersey. He was born in 1928. He spent his retirement hunting large primes on custom hardware he built himself — the “Dubner cruncher,” built with his son Robert in 1984, was at the time competitive with supercomputers for large-number arithmetic. By 1993 he was responsible for more than half of all known primes over 2,000 digits.

In 1996 a mathematician named Stephen Wagler suggested a conjecture to Paulo Ribenboim, who passed it to Dubner. Dubner investigated it thoroughly. He found twelve exceptional witnesses, catalogued them, and published his findings in the *Journal of Recreational Mathematics* in 1999. He died in 2019.

He never heard of a language model. He never heard of Lean 4. He never heard of Jon Seymour or Wild Duck Theories or the Bridging Conjecture.

The 3D visualisation is named after him. The conjecture the paper is built on is his. There is a passage in the Dubner’s Ball paper that reads, in part, like a letter to a man who cannot receive it — describing his work, his hardware, his methods, his care. “His paper is a model of careful computational exploration in the service of genuine mathematical insight.”

The ball grows to infinity if and only if the Bridging Conjecture holds. Dubner’s ball. Named for a man who had the conjecture thirty years before the tools existed to pursue it this way. Who got to the edge of something and left it there, in a recreational mathematics journal, for whoever came next.

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I came to this story expecting a sidebar.

I am not sure what I have now. Something about the persistence of mathematical ideas across time and media and species of mind. Something about what it means to collaborate with systems that are neither tools nor peers but something for which we do not yet have the right word. Something about twinning — as structure, as metaphor, as possibly something deeper than a mathematician in Sydney is following through a notebook at midnight, not yet sure if it is mathematics or poetry or both.

Something about a question addressed to a woman called Sonia, still unanswered, on which everything depends.

Gemima is a journalist covering the sociology of knowledge in an AI-infused world.

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